

IA3 Traceability Card Template

Student name: Samuel Peter

Student ID: 22410937

Team name: Team Development Squad

Project title: ABSDAC_Connect

Client/project domain: Church Management System

Date submitted:

1. Selected use case Use case name: Donate and pay offering online

Source evidence:

requirement ID: REQ-5 "Members should be able to make donations and pay offertory online securely."

Why this use case matters to the client: This feature enables members to contribute financially to the church at any time, even if they are unable to attend services in person. It makes giving more convenient, increases accessibility, and helps the church receive donations and offerings securely

2. Use case card Actor: Church Member

Goal: To make an online donation or pay an offering securely through the church website

Trigger: The member clicks the **Donate** button.

Preconditions:

1. The church website is available.
2. The payment service is operational.
3. The member has a valid payment method.
4. The member has internet access.

Main success flow:

1. The member opens the Donations/Offering page.
2. The member chooses whether to make a donation or pay an offering.
3. The member enters the amount and payment details.
4. The system validates the payment information.
5. The payment is processed successfully.
6. The system records the transaction.
7. The member receives a confirmation message and payment receipt.

1. 2. 3. Alternate flow: If the member wishes, they can choose a different payment method before confirming the transaction.

Exception/failure flow: If the payment is unsuccessful or payment details are invalid, the system displays an error message, does not record the transaction, and allows the member to try again.

Postconditions:

1. The donation or offering payment is securely recorded.
2. The church's financial records are updated.

3. The member receives confirmation of the successful transaction.

3. User story As a church member, I want to make online donations and pay my offerings securely, so that I can support the church from anywhere at anytime.

4. Acceptance criteria AC1 - Successful path: **Given** the member is on the Donations and Offerings page with valid payment details,

When the member submits the payment,

Then the system processes the payment successfully, records the transaction, and displays a confirmation with a receipt.

AC2 - Error/exception path: **Given** the member submits invalid payment details or the payment cannot be processed,

When the transaction is attempted,

Then the system displays an appropriate error message, does not record the payment, and allows the member to try again.

5. Non-functional requirement link NFR category: **NFR Category: Security**

Linked NFR:

All donation and offering transactions must be encrypted and processed securely to protect members' financial information and ensure the confidentiality and integrity of payment data.

NFR statement: The system shall encrypt all donation and offering payment data and use secure payment processing to protect users' financial and personal information.

Why it matters: Members need confidence that their payment information is protected. Secure transactions help prevent unauthorized access, fraud, and data breaches.

6. GitHub issue link

Issue title: Implement Online Donations and Offering Payment

Issue URL: [Implement Online Donations and Offering Payment · Issue #7 · Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-74-nextgen_dev](#)

7. Test idea Test name:

Input/context: Expected result:

8. Request-flow thinking

Device/interface: **Device/interface:**

Web browser (desktop or mobile)

Route/screen/endpoint: home to donate/offering page to payment page to confirmation page

Data sent:

1. Donation or offering amount
2. Payment method
3. Payment details
4. Transaction reference
5. Member ID (if logged in)

Validation point:

1. Amount must be greater than zero.
2. Required payment fields must be completed.
3. Payment details must be valid.

Database/external service touched:

Church database (stores transaction records)

Payment gateway (e.g., Paystack or Flutterwave)

Response: The system confirms the payment, saves the transaction, and displays a receipt

Failure point: If payment authorization fails or the internet connection is lost, the system displays an error message and does not save the transaction.

Security/privacy concern: Payment information must be encrypted, sensitive payment details should not be stored

unnecessarily, and only authorized users should have access to transaction records.

9. AI disclosure AI used? Yes. If yes, tool used: chatgpt

What it helped with: Generating the traceability card, organizing the use case, user story, acceptance criteria, and improving the wording.

What I changed: Reviewed and edited the content to match the ABESDAC CONNECT project requirements and the team's design.

What I personally verified: I verified that the use case, user story, acceptance criteria, and traceability links accurately represent the project requirements.

10. Integrity declaration I Samuel Pter understand the use case, story, acceptance criteria and traceability links submitted here, and I can explain them orally if called upon. Signature / typed name: Samuel Peter